MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

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VECTORS IN THREE-DIMENSIONS

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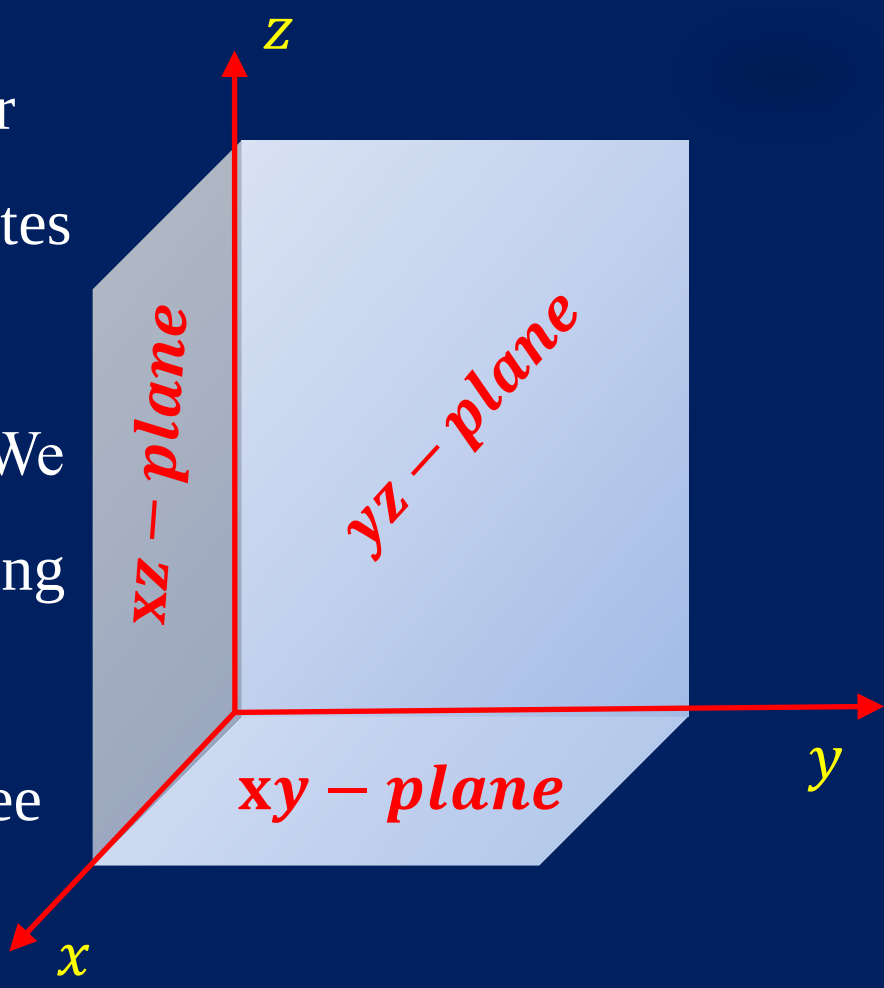
INTRODUCTION

The concepts discussed for vectors in the plane can be generalized for vectors in 3-dimensional space.

To discuss vectors in space, we use ordered triples and a 3-dimensional coordinate systems. An ordered triple (x, y, z) is a set of $\{x, y, z\}$ of three numbers in which x is a 1st number of the set, y is the 2nd and z the 3rd .

Two ordered triples (x_1, y_1, z_1) and (x_2, y_2, z_2) are equal, and we write $(x_1, y_1, z_1) = (x_2, y_2, z_2) \Leftrightarrow x_1 = x_2, y_1 = y_2$ and $z_1 = z_2$. The totality of order triples is denoted by $\mathbb{R}^3$.

To Obtain a rectangular coordinate system in 3-dimensional, we choose a fixed-point O (the origin) and consider three mutually perpendicular coordinates lines (the $(x, y, and\ z-axes)$) with common origin 'O' as shown in Fig. We may regard the $y, and\ z-axes$ as lying in the plane of the paper and x-axis as projecting out from the paper. The three coordinates lines determine three coordinates planes: $xy-plane$, the $yz-plane$ and the $xz-plane$.



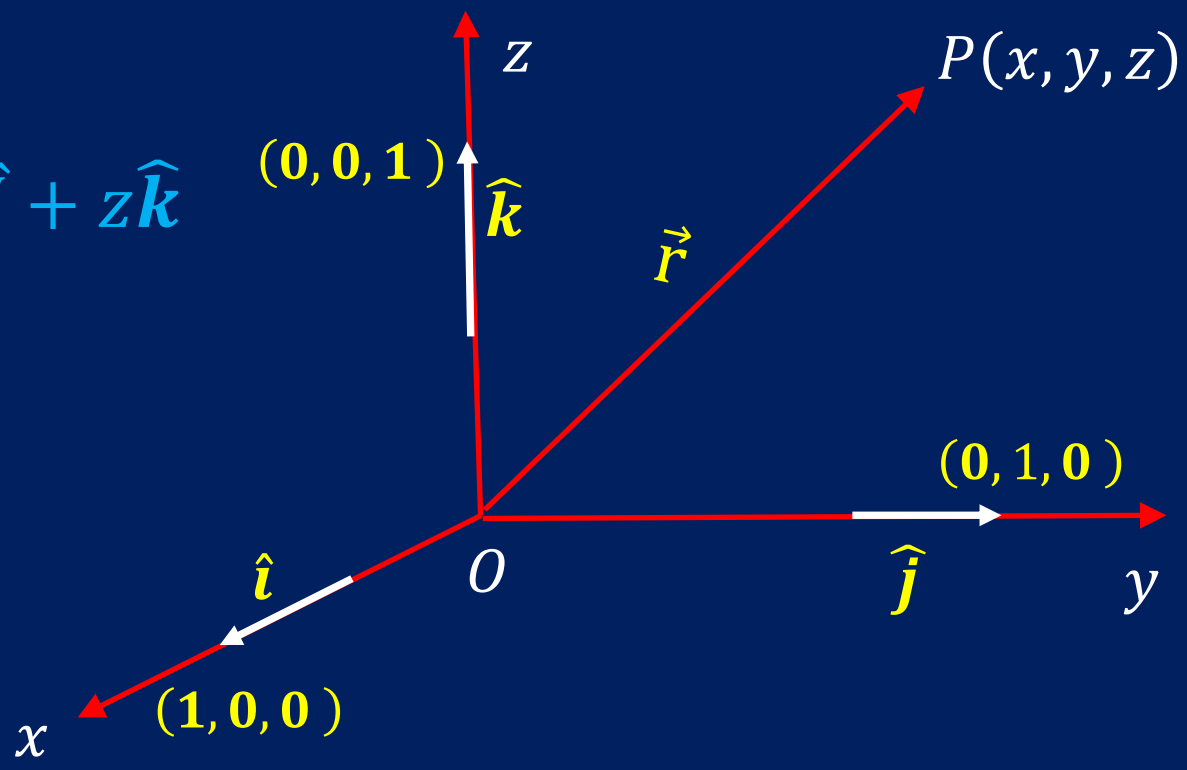
VECTORS IN SPACE

The vectors represented by the directed line segments from the origin to the points $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ are the basic vectors (see Fig).

We denote them by $\hat{\imath}, \hat{\jmath}$ *and* $\hat{k}$.

The position vector from the origin 'O' to the typical point $P(x, y, z)$ is

$$\vec{r} = \overrightarrow{OP} = x\hat{\imath} + y\hat{\jmath} + z\hat{\boldsymbol{k}}$$



ADDITION & SUBTRACTION FOR VECTORS IN SPACE

Algebraic Definition

For any vectors

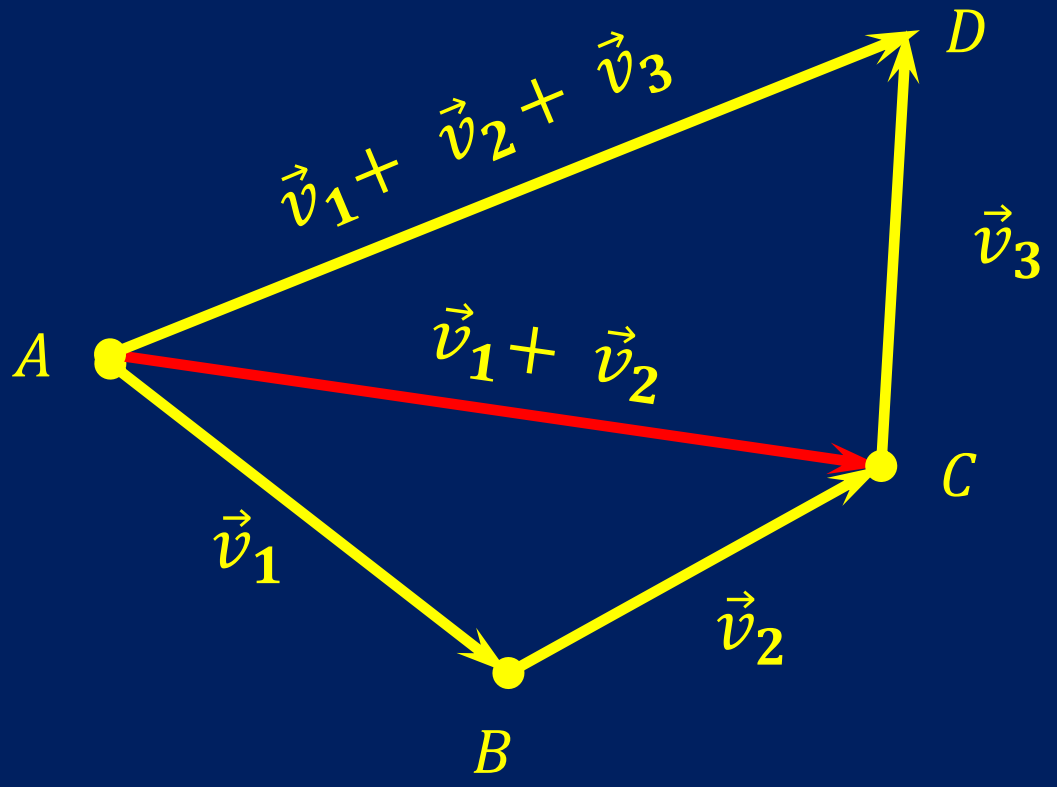
$\vec{A} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$ and

$\vec{B} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$ then

$$\vec{A} + \vec{B} = (a_1 + b_1)\,\hat{\imath} + (a_2 + b_2)\hat{\jmath} + (a_3 + b_3)\hat{k}$$

$$\vec{A} - \vec{B} = (a_1 - b_1)\,\hat{\imath} + (a_2 - b_2)\hat{\jmath} + (a_3 - b_3)\hat{k}$$

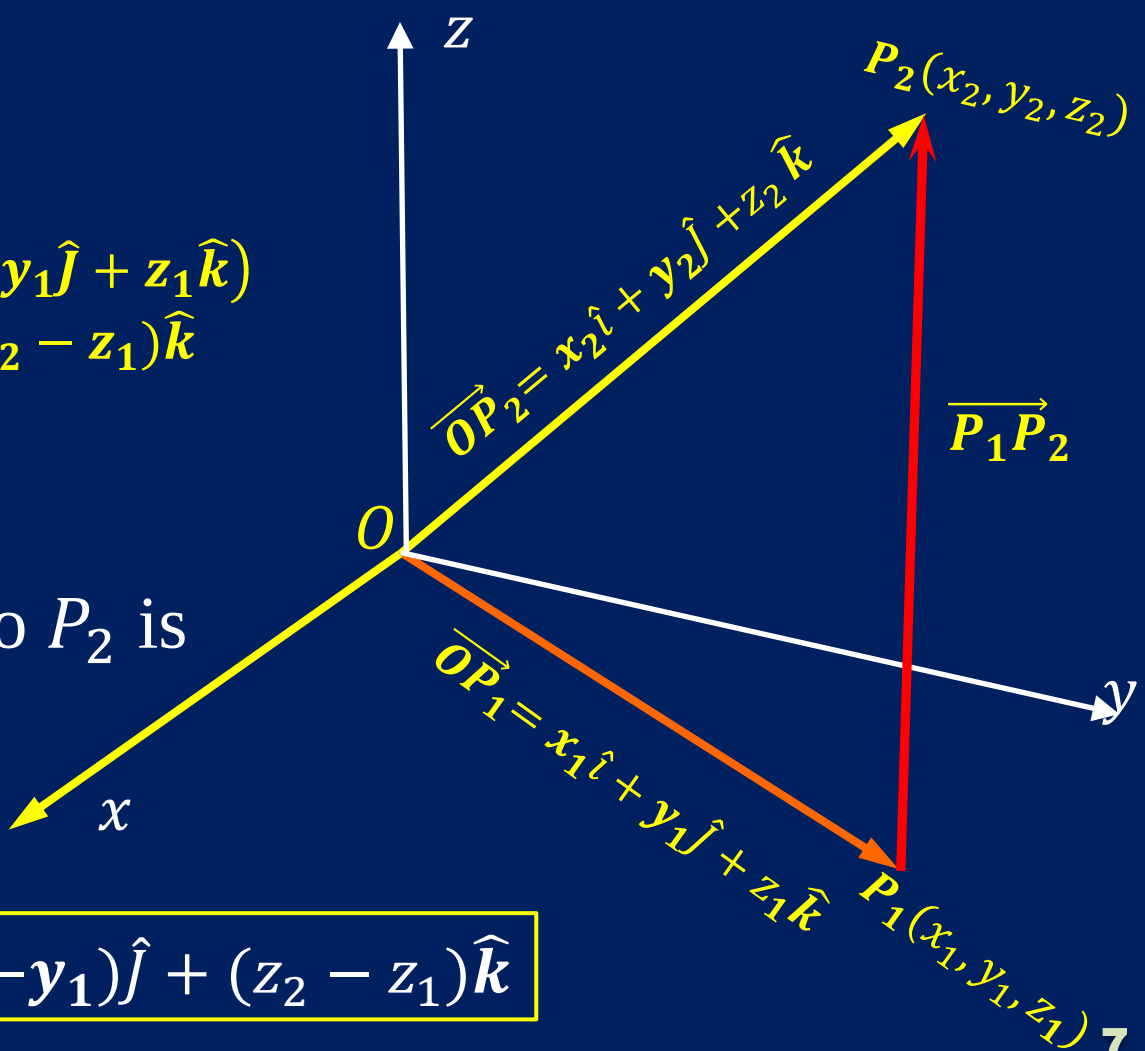
Geometrically



THE VECTORS b/w TWO POINTS

We can express the vectors $\overrightarrow{P_1P_2}$ from the point $P_1(x_1, y_1, z_1)$ to a point $P_2(x_2, y_2, z_2)$ in terms of the coordinates of P_1 and P_2

$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$$
$$= (x_2\hat{\imath} + y_2\hat{J} + z_2\hat{k}) - (x_1\hat{\imath} + y_1\hat{J} + z_1\hat{k})$$
$$= (x_2 - x_1)\hat{\imath} + (y_2 - y_1)\hat{J} + (z_2 - z_1)\hat{k}$$



So the vectors from P_1 to P_2 is

$$\overrightarrow{P_1P_2} = (x_2 - x_1)\hat{\imath} + (y_2 - y_1)\hat{J} + (z_2 - z_1)\hat{k}$$

MAGNITUDE OF VECTOR

The magnitude or length of $\vec{v} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ is

$$|\vec{v}| = \left|x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right| = \sqrt{x^2 + y^2 + z^2}$$

SCALAR MULTIPLICATION

If C is a scalar and $\vec{v} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ is a vector, then

$$c\vec{v} = \mathrm{c}\left(x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right) = (cx)\hat{\imath} + (cy)\hat{\jmath} + (cz)\hat{k}$$

Example:

The length of $\vec{v} = \hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ is

$$|\vec{v}| = \sqrt{(1)^2+(-2)^2+(3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$

REMARK

If we multiply $\vec{v} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ by a scalar c, the length of $c\vec{v}$ is |c| times the length of $\vec{v}$, as in the plane.

$$c\vec{v} = \mathrm{c}\left(x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right) = (cx)\hat{\imath} + (cy)\hat{\jmath} + (cz)\hat{k}$$

$$|\, c\vec{v}\,| = |(cx)\hat{\imath} + (cy)\,\hat{\jmath} + (cz)\hat{k}|$$

$$= \sqrt{c^2x^2 + c^2y^2 + c^2z^2}$$

$$= \sqrt{c^2(x^2 + y^2 + z^2)}$$

$$= \sqrt{c^2}\sqrt{(x^2+y^2 + z^2)}$$

$$= |\, c\,|\,|\,\vec{v}\,|$$

For Example

If $c = 2$ and $\vec{v} = \hat{\imath} - 2\hat{\jmath} + 3\hat{\boldsymbol{k}}$

then length of $c\vec{v} = 2\left(\hat{\imath} - 2\hat{\jmath} + 3\hat{k}\right) = 2\hat{\imath} - 4\hat{\jmath} + 6\hat{k}$ is

$$|\, c\vec{v}\,| = \sqrt{(2)^2 + (-4)^2 + (6)^2} = \sqrt{4 + 16 + 36} = \sqrt{56}$$

$$= 2\sqrt{14} = 2\,|\vec{v}\,|$$

ZERO VECTOR

The zero vector in space is the vector

$$\vec{0} = 0\hat{\imath} + 0\hat{\jmath} + 0\hat{k}$$

0 has zero length and no direction

UNIT VECTORS

A unit vector in space is a vector of length 1. The basic vectors are unit vectors b/c

$$|\hat{\imath}| = |1\,\hat{\imath} + 0\,\hat{\jmath} + 0\hat{k}| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

$$|\hat{\jmath}| = |0\hat{\imath} + 1\,\hat{\jmath} + 0\hat{k}| = \sqrt{0^2 + 1^2 + 0^2} = 1$$

$$|\hat{k}| = |0\hat{\imath} + 0\,\hat{\jmath} + 1\hat{k}| = \sqrt{0^2 + 0^2 + 1^2} = 1$$

MAGNITUDE AND DIRECTION

If $\vec{v} \neq 0$, then is

1. $\frac{\vec{v}}{|\vec{v}|}$ is a unit vector in the direction of $\vec{v}$.
2. The equation $\vec{v} = |\vec{v}| \left(\frac{\vec{v}}{|\vec{v}|}\right)$ express $\vec{v}$ as a product of its magnitude and direction

EXAMPLE;

Express $\vec{v} = \hat{\imath} - 2\hat{\jmath} + 3\hat{\boldsymbol{k}}$ as a product of its magnitude and direction

Solution:

Length of $\vec{v}$

$$|\vec{v}| = \left|\hat{\imath} - 2\hat{\jmath} + 3\hat{k}\right| = \sqrt{1^2 + (-2)^2 + (3)^2}$$

$$= \sqrt{1 + 4 + 9} = \sqrt{14}$$

Direction of $\vec{v}$

$$\frac{\vec{v}}{|\vec{v}|} = \frac{\hat{\imath} - 2\hat{\jmath} + 3\hat{k}}{\sqrt{14}} = \frac{1}{\sqrt{14}}\hat{\imath} - \frac{2}{\sqrt{14}}\hat{\jmath} + \frac{3}{\sqrt{14}}\hat{k}$$

Now product of magnitude and direction of $\vec{v}$ is

so $\vec{v} = \sqrt{14}\left(\frac{1}{\sqrt{14}}\hat{\imath} - \frac{2}{\sqrt{14}}\hat{\jmath} + \frac{3}{\sqrt{14}}\hat{k}\right)$

Length

Direction

EXAMPLE:

Express $\vec{a} = (3, -4, 2)$ in terms of $\hat{\imath}, \hat{\jmath}$ *and* $\hat{\boldsymbol{k}}$ and find the unit vector $\vec{u}$ that has the same direction as $\vec{a}$.

Solution:

We may write

$$\vec{a} = (3, -4, 2) = 3\hat{\imath} - 4\hat{\jmath} + 2\hat{\boldsymbol{k}}$$

The magnitude of $\vec{a} = |\vec{a}| = \sqrt{9 + 16 + 4} = \sqrt{29}$

The unit vector $\vec{u}$ having the same direction as $\vec{a}$ is

$$\vec{u} = \frac{\vec{a}}{|\vec{a}|} = \frac{3\hat{\imath} - 4\hat{\jmath} + 2\hat{k}}{\sqrt{29}} = \frac{3}{\sqrt{29}}\hat{\imath} - \frac{4}{\sqrt{29}}\hat{\jmath} + \frac{2}{\sqrt{29}}\hat{k}$$

EXAMPLE:

Find a unit vector $\vec{u}$ in the direction of the vector from $P_1(1,0,1)$ to a point $P_2(3,2,0)$.

Solution:

$$\overrightarrow{P_1P_2} = (3-1)\hat{\imath} + (2-0)\hat{\jmath} + (0-1)\hat{k} = 2\hat{\imath} + 2\hat{\jmath} - \hat{k}$$

$$\left|\overrightarrow{P_1P_2}\right| = \sqrt{4+4+1} = \sqrt{9} = 3$$

The unit vector $\vec{u}$ is

$$\vec{u} = \frac{\overrightarrow{P_1P_2}}{\left|\overrightarrow{P_1P_2}\right|} = \frac{2\hat{\imath} + 2\hat{\jmath} - \hat{k}}{3} = \frac{2}{3}\hat{\imath} + \frac{2}{3}\hat{\jmath} - \frac{1}{3}\hat{k}$$

EXAMPLE: Find a vector 6 units long in the direction of

$$\vec{u} = 2\hat{\imath} + 2\hat{\jmath} - \hat{k}$$

Solution Try yourself

DISTANCE IN SPACE

The distance between two points P_1 **and** P_2 and in space is the length of $\overrightarrow{P_1P_2}$.

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ then

$$\left|\overrightarrow{P_1P_2}\right| = \sqrt{(\boldsymbol{x_2 - x_1})^2 + (\boldsymbol{y_2 - y_1})^2 + (z_2 - z_1)^2}$$

For Example:

Find the distance between the Points $A(2,1,5)$ and $B(-2,3,0)$.

$$\left|\overrightarrow{AB}\right| = \sqrt{(-2-2)^2 + (3-1)^2 + (0-5)^2}$$

$$= \sqrt{16 + 4 + 25} = \sqrt{45} = 3\sqrt{5}$$

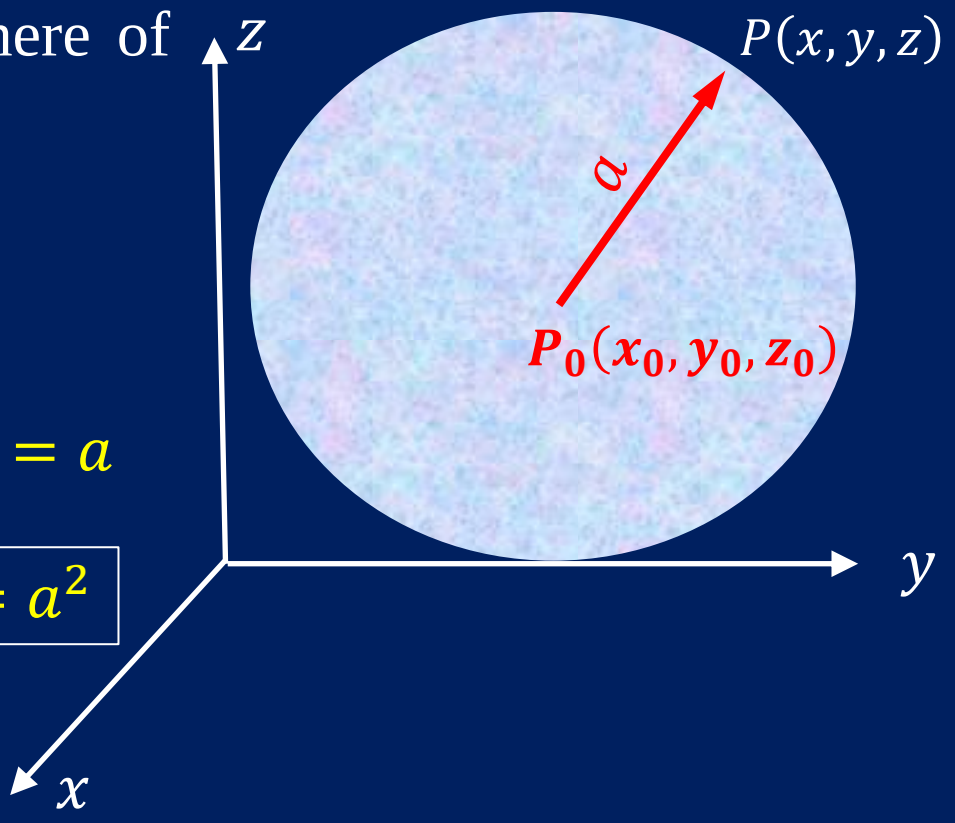
EQUATION OF SPHERES

To write equations for spheres in space we use distance formula in space.

A Point $P(x, y, z)$ lies on the sphere of radius a centered at $P_0(x_0, y_0, z_0)$

So

$$|\overrightarrow{P_0P}| = a$$



$$\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2} = a$$

$$(x-x_0)^2+(y-y_0)^2+(z-z_0)^2 = a^2$$

Above equation is called the standard equation for the sphere of radius a centered at (x_0, y_0, z_0)

EXAMPLE:

Find the center and radius of the sphere $x^2 + y^2 + z^2 - 6x + 8y + 4z + 4 = 0$

Solution:

We find the center and radius of a sphere the way we find the center and radius of a circle. Complete the squares on the x, y, z terms as necessary and write each quadratic as a squared linear expression. Then form the equation in standard form, read oof the center and radius.

For the sphere here, we have

$$x^2 + y^2 + z^2 - 6x + 8y + 4z + 4 = 0$$

$$\Rightarrow (x^2 - 6x) + (y^2 + 8y) + (z^2 + 4z) + 4 = 0$$

$$\Rightarrow x^2 - 6x + 9 + y^2 + 8y + 16 + z^2 + 4z + 4 = -4 + 9 + 16 + 4$$

$$\Rightarrow (x - 3)^2 + (y + 4)^2 + (z + 2)^2 = 25$$

So the center is $(3, -4, -2)$ and radius is 5

EXERCISE

Q#1 Express each vector as a Product of its length and direction.

a) $2\hat{i} + \hat{j} - 2\hat{k}$ (b) $3\hat{i} - 6\hat{j} + 2\hat{k}$ (c) $\hat{i} + 4\hat{j} - 8\hat{k}$

d) $9\hat{i} - 2\hat{j} + 6\hat{k}$ (e) $5\hat{k}$ (f) $-4\hat{j}$

g) $\frac{3}{5}\hat{i} + \frac{4}{5}\hat{k}$ (h) $\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{k}$ (i) $\frac{1}{\sqrt{6}}\hat{i} - \frac{1}{\sqrt{6}}\hat{j} - \frac{1}{\sqrt{6}}\hat{k}$

j) $\frac{\hat{i}}{\sqrt{3}} + \frac{\hat{j}}{\sqrt{3}} + \frac{\hat{k}}{\sqrt{3}}$

Q#2 Find a vector of magnitude 7 in the direction of $\vec{A} = 12\hat{i} - 5\hat{k}$.

Q#3 Find a vector $\sqrt{5}$ units long in the direction of $\vec{A} = \hat{i} + \hat{j} + \hat{k}$

Q#4 Find a vector 5 units long in the direction opposite to the direction of $\vec{A} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

Q#5 Find a vector of magnitude '3' in the direction opposite to the direction of

$$\vec{A} = \frac{1}{2}\hat{i} - \frac{1}{2}\hat{j} - \frac{1}{2}\hat{k}$$

Q#6 If $\overrightarrow{AB} = \hat{i} + 4\hat{j} - 2\hat{k}$ and B is the point (5, 1, 3), find A.

Q#7 If $\overrightarrow{AB} = -7\hat{i} + 3\hat{j} + 8\hat{k}$ and A is the Point $(-2, -3, 6)$, find B.

Q#8 Find (a) the distance b/w Points P_1 and P_2

(b) the direction of $\overrightarrow{P_1P_2}$

i) $P_1(1, 1, 1)$, $P_2(3, 3, 0)$

ii) $P_1(-1, 1, 5)$, $P_2(2, 5, 0)$

iii) $P_1(1, 4, 5)$, $P_2(4, -2, 7)$

iv) $P_1(3, 4, 5)$, $P_2(2, 3, 4)$

v) $P_1(0, 0, 0)$, $P_2(2, -2, -2)$

vi) $P_1(5, 3, -2)$, $P_2(0, 0, 0)$

Q#9 Find the unit vector that has the same direction as $\vec{a}$

(a) $\vec{a} = 2(-2, 5, -1)$

(b) $\vec{b} = 3\hat{i} - 7\hat{j} + 2\hat{k}$

Q#10 Find an equation of the sphere whose centers and radii are given

a) center $(1, 2, 3)$ radius $\sqrt{14}$

b) center $(0, -1, 5)$ radius 2

c) center $(-2, 0, 0)$ radius $\sqrt{3}$

d) center $(0, -7, 0)$ radius 7

e) center $(3, -1, 2)$ and radius 2

f) center $(4, -5, 1)$ and radius 5

g) center $(-5, 0, 1)$ and radius $1/2$

h) center $(0, -3, -6)$ and radius $\sqrt{3}$

Q # 11 Find the center and radius of spheres

a) $x^2 + y^2 + z^2 + 4x - 4z = 0$

b) $x^2 + y^2 + z^2 - 6y + 8z = 0$

c) $2x^2 + 2y^2 + 2z^2 + x + y + z = 9$

d) $3x^2 + 3y^2 + 3z^2 + 2y - 2z = 9$

e) $x^2 + y^2 + z^2 + 4x - 2y + 2z + 2 = 0$

f) $x^2+y^2+z^2-6x-10y+6z+34=0$

g) $x^2+y^2+z^2-8x+8z+16=0$.

h) $4x^2+4y^2+4z^2-4x+8y-3=0$

i) $x^2+y^2+z^2+4y=0$

j) $x^2+y^2+z^2-z=0$

Q#12 Find the centeres and radii of the spheres

a) $(x+2)^2+y^2+(z-2)^2=8$

b) $(x+\frac{1}{2})^2+(y+\frac{1}{2})^2+(z+\frac{1}{2})^2=\frac{21}{4}$

c) $(x-\sqrt{2})^2+(y-\sqrt{2})^2+(z+\sqrt{2})^2=2$